

CODES AND CIPHERS



A CODE IS ANY SYSTEM of prearranged symbols, words, or numbers that is used in communication. For example, the flags that are used to send messages at sea are a naval code. We use codes to simplify, organize, and communicate complex information, for instance, in dialling and postal codes, or bar codes that describe goods in a way that machines can read. Not all codes have an everyday use. Ciphers (secret codes) hide the true meaning of a message. Banks use them to keep financial deals private, and spies or criminals to avoid capture.

Ciphers

In a cipher, each letter is represented by a different letter or symbol. For instance, it is easy to encipher a message by jumbling the alphabet, changing C into M and M to C. It is easy to break such a simple cipher, but computers can create ciphers that are impossible to read without the key (a long number that unlocks the meaning).



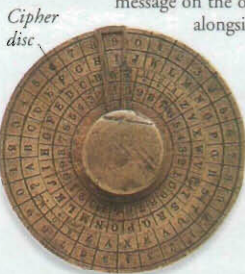
Spies

A spy is a secret agent, who collects information for a government or organization. A spy's work often involves stealing the secrets of rival governments. Spies use ciphers to scramble data when they send it to their employers.

Spies codesheet

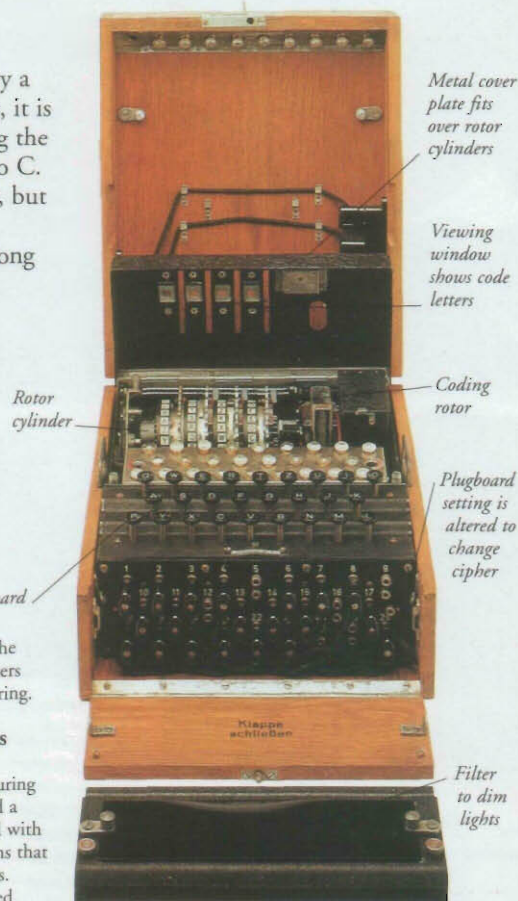
Cipher discs

These devices create ciphers by replacing letters of the message on the outer ring with the letters alongside them on the inner ring.



Cipher machines

The Enigma cipher machine was used during World War II. It had a typewriter keyboard with electrical connections that scrambled the letters. Each letter was coded separately, making the cipher hard to break.



German Enigma cipher machine, World War II

Smoke signals

Fire beacons and smoke codes were used to send signals by the people of ancient China, Egypt, and Greece. Native Americans, such as the Cheyenne, Comanche, and Sioux, communicated over distances using smoke signals, shaping smoke with an animal hide or blanket. There were a few generally understood signals – two puffs meant “all’s well” – but each group also had secret codes which they shared only with people they wanted to read the messages.



Frederic Remington, *Smoke Signals*

Uses of code

Codes make messages quicker to send. They have been used for many reasons. Sailors, for example, used flag codes to communicate for more than 1,000 years. By flying the three flags standing for the letters NKA, a warship could send a message meaning “I have not sighted any vessels since leaving my last port”. A code book carried on every ship translated the codes.

Computer codes

Special codes are used to program information inside computers, where letters and punctuation marks are represented by binary numbers. Ciphers can also be used to protect e-mail (mail sent between computers), so that it can only be understood by the sender and the addressee.



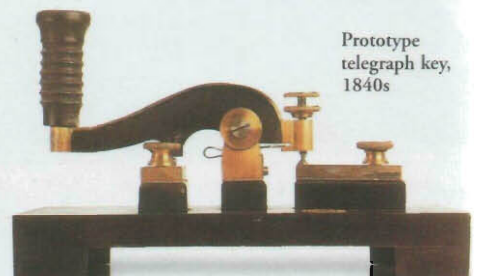
E-mail can be encrypted so that only the addressee, who holds a secret “key” (a long number), can read it.

The alphabet in Morse code

A	• –	N	– •
B	– • • •	O	– – –
C	– • – •	P	– • – •
D	– • •	Q	– – • –
E	•	R	• – •
F	• • – •	S	• • •
G	– – •	T	–
H	• • • •	U	• • –
I	• •	V	• • • –
J	• – –	W	• – –
K	– • –	X	– • • –
L	• – • •	Y	– • – –
M	– –	Z	– • • •

Morse code

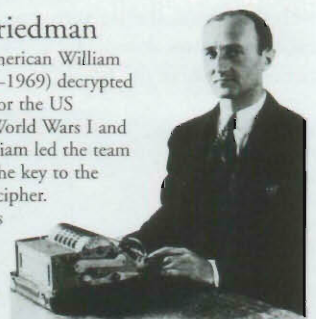
The telegraph was invented in the 19th century; it used electricity to send messages quickly over long distances for the first time. The system could not transmit speech, so to communicate operators used an alphabetic code devised by US artist Samuel Morse (1791–1872). Letters were represented by dots and dashes (long and short pulses of power). Operators tapped a key to turn the electric current in the telegraph wires on and off. Morse code is still in use.



Prototype telegraph key, 1840s

William Friedman

Russian-born American William Friedman (1891–1969) decrypted secret messages for the US government in World Wars I and II. In 1940, William led the team that discovered the key to the Japanese Purple cipher. A message in this cipher warned of the Pearl Harbor attack.



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